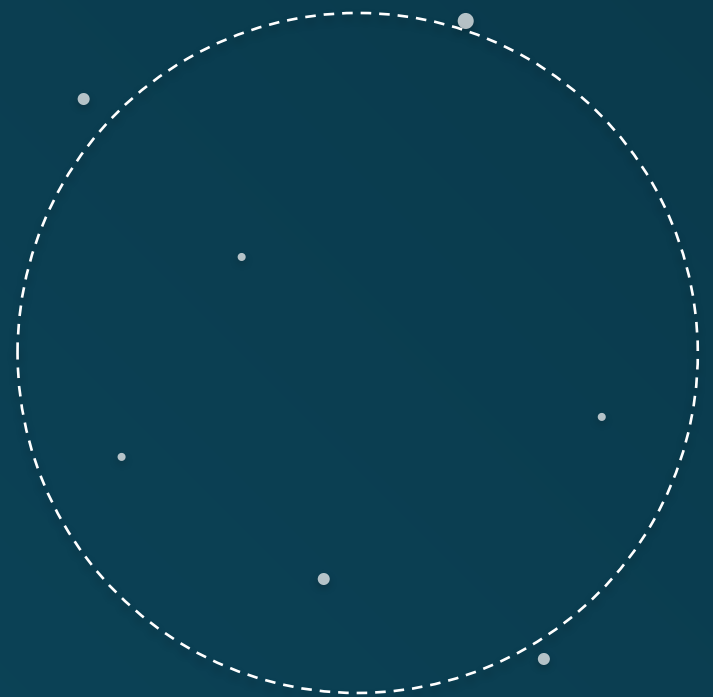


TOPIC 15

Ideal Gases

理想气体



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

15.1 The mole 摩尔

- Amount of substance is an SI base quantity with base unit mol 物质的量是基本量
- One mole contains the Avogadro constant $N_A = 6.02 \times 10^{23}$ particles 阿伏伽德罗常数

15.2 Equation of state 状态方程

- An ideal gas obeys $pV \propto T$ (T in kelvin) 理想气体定义
- Recall and use $pV = nRT$ and $pV = NkT$; $k = R/N_A$ 两种形式的状态方程

15.3 Kinetic theory of gases 分子动理论

- State the basic assumptions of the kinetic theory 动理论基本假设
- Explain pressure from molecular motion; derive and use $p = \frac{1}{3} \rho \langle c^2 \rangle$ 压强微观推导
- Relate mean kinetic energy to thermodynamic temperature 平均动能与温度

The Mole

摩尔与阿伏伽德罗常数

Amount of substance is an SI base quantity; its base unit is the **mole (mol)**.

物质的量是 SI 基本量，单位摩尔。

One mole of any substance contains a number of particles equal to the **Avogadro constant** .

1 摩尔物质含有 N_A 个粒子。

$$n = \frac{N}{N_A} = \frac{m}{M}$$

N 粒子数 · m 质量 · M 摩尔质量

- Molar mass 摩尔质量 : mass of 1 mol, in g mol^{-1} (数值 $\approx$ 相对原子 / 分子质量)
- Counting by moles links the microscopic (molecules) to the macroscopic (grams) 摩尔是微观与宏观的桥梁

Equation of State

理想气体状态方程

An **ideal gas** is a gas that obeys , where T is the **thermodynamic temperature**.

理想气体：任意状态下 pV 与热力学温度成正比的气体。

$$pV = nRT \quad pV = NkT$$

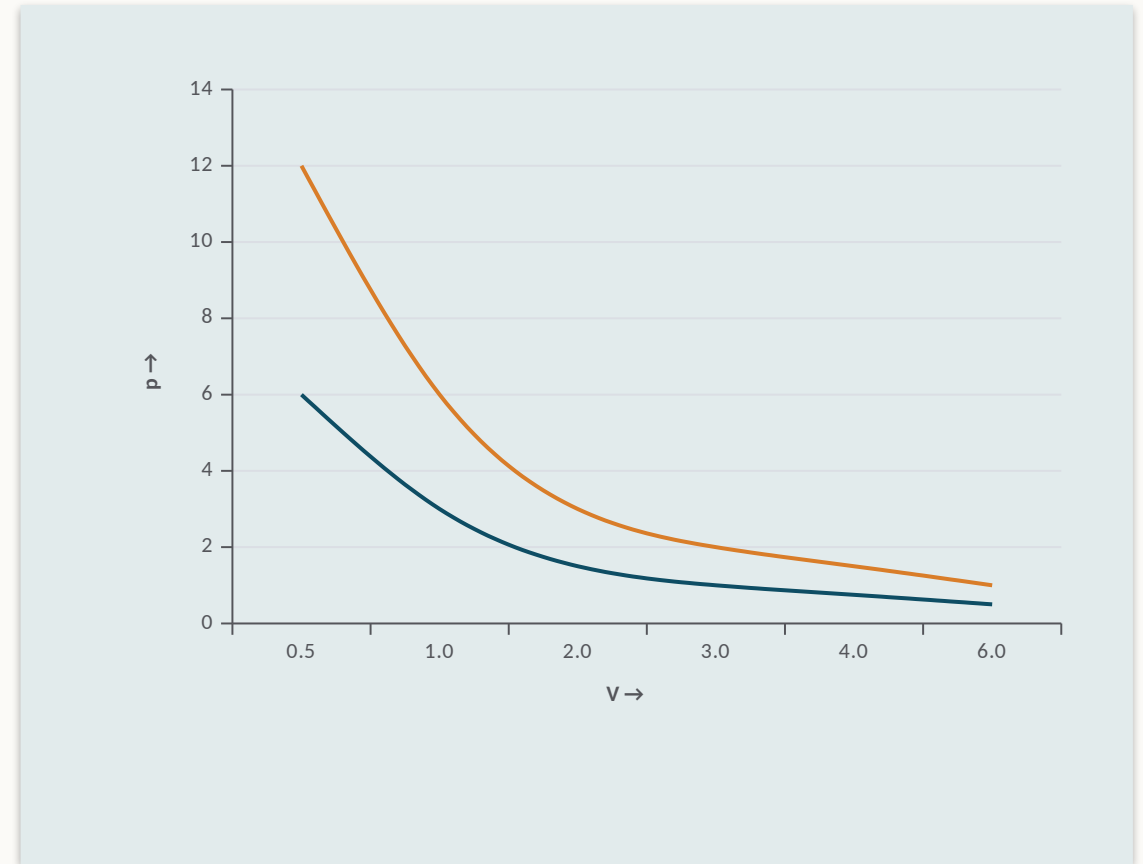
n 摩尔数 · N 分子数

- molar gas constant 普适气体常量
- **Boltzmann constant** 玻尔兹曼常数
- Two forms are equivalent: 两式等价
- T **must** be in kelvin 温度必须用开尔文

Gas Laws in Graphs

气体实验定律与图像

- **Boyle's law** 玻意耳定律 : $pV = \text{constant}$ at fixed T (p - V 双曲线; T 越高等温线越靠外)
- **Charles' law** 查理定律 : $V \propto T$ at fixed p (V - T 过原点直线, T 用 K)
- **Pressure law** 压强定律 : $p \propto T$ at fixed V
- All three combine into 三定律统一为状态方程
- Two states of a fixed mass: 两状态联立式



p-V isotherms of a fixed mass of gas 等温线

Kinetic Theory Assumptions

分子动理论基本假设

1. A gas consists of a **very large number** of molecules in continuous **random motion** 大量分子无规则运动
2. The volume of the molecules is **negligible** compared with the volume of the gas 分子本身体积可忽略
3. Intermolecular forces are **negligible** except during collisions 除碰撞瞬间无分子间力
4. All collisions (molecule-molecule and molecule-wall) are **perfectly elastic** 碰撞完全弹性
5. The time of collisions is negligible compared with the time between collisions 碰撞时间极短



Random motion & elastic collisions with walls 分子无规则运动并与器壁弹性碰撞

Pressure from Molecular Motion

压强的微观推导

- One molecule hits a wall: momentum change per collision 单次碰撞动量改变
- Collision rate with that wall: 往返时间 $2L/v_x$
- Force = rate of change of momentum; average over N molecules with (三个方向均等)
- Result 结论:

$$p = \frac{1}{3} \rho \langle c^2 \rangle = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

$\rho = Nm/V$ 气体密度 · $\langle c^2 \rangle$ mean-square speed 方均速率

Mean Kinetic Energy & r.m.s. Speed

平均动能与方均根速率

- Combine with : 联立两式

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} k T$$

average KE per molecule 单分子平均动能

- **Temperature is a measure of the mean kinetic energy** of molecules 温度是分子平均动能的量度
- Root-mean-square speed 方均根速率 :
- For one mole: total KE 每摩尔总动能
- Ideal gas internal energy = kinetic energy only (no intermolecular PE) 理想气体内能仅有动能

Worked Example 15A

例题解析

WORKED EXAMPLE 15A

1 The molar mass of neon is 20.0 g. Calculate the number of atoms in a 2.5 g sample.

2 A cylinder contains oxygen molecules. The mass of 1 mol of oxygen atoms is 16.0 g. Determine (a) the amount in mol; (b) the mass of oxygen.

注意区分 "原子" 与 "分子" 的摩尔质量 (O₂ 要 ×2)。

SOLUTION 解答

Step 1 1 20.0 g contains atoms $\Rightarrow$ 2.5 g contains atoms.

Step 2 2 (a)

Step 3 (b) Oxygen molecules are O₂: molar mass ; mass

Answer 1 atoms 2 (a) 2.51 mol (b) 80.3 g

Worked Example 15B

例题解析

WORKED EXAMPLE 15B

1 Calculate the volume occupied by 1 mole of air at s.t.p. (273 K, ;).

2 A syringe contains of helium at 20 °C and . The temperature rises to 400 °C and the pressure to . Find the new volume.

摩尔体积 $\approx 22.4 \text{ dm}^3$ 值得记住；两状态题先把温度换成 K。

SOLUTION 解答

Step 1 **1**

Step 2 **2** Convert temperatures: , .

Step 3 Two-state equation:

Answer **1** (22.4 dm³) **2**

Worked Example 15C

例题解析

WORKED EXAMPLE 15C

The speeds of seven molecules are numerically 2, 4, 6, 8, 10, 12 and 14 units. Determine: (a) the mean speed $\langle c \rangle$; (b) the mean speed squared $\langle c \rangle^2$; (c) the mean-square speed $\langle c^2 \rangle$; (d) the r.m.s. speed.

经典易混点: $\langle c \rangle^2 \neq \langle c^2 \rangle$, 注意 "先平均还是先平方"。

SOLUTION 解答

Step 1 (a) units

Step 2 (b) units²

Step 3 (c) units²

Step 4 (d) units

Answer $\langle c \rangle = 8.0$ · $\langle c \rangle^2 = 64$ · $\langle c^2 \rangle = 80$ · rms = 8.9 units

Worked Examples 15D & 15E

例题解析

WORKED EXAMPLE 15D

Calculate the total kinetic energy of the molecules in one mole of an ideal gas at 273 K. ()

1. Per molecule: ; per mole:
- 2.

Answer

WORKED EXAMPLE 15E

The mass of a nitrogen molecule is . Determine the r.m.s. speed of nitrogen molecules at 27 °C.

1. ;
- 2.

Answer

Class Practice

随堂练习

- Q1** The mass of 1 mol of nitrogen atoms is 14.0 g. Determine (a) the mass of one nitrogen atom in kg; (b) the number of nitrogen molecules (N_2) in a cylinder containing 290 g of nitrogen.
- Q2** A fixed mass of gas is at 27 °C and pressure . It is heated at constant volume to 127 °C. Calculate the new pressure. 等容升温求压强。
- Q3** Determine (a) the average kinetic energy of a molecule in an ideal gas at 260 °C; (b) the r.m.s. speed of neon-20 molecules at 50 °C (mass of one molecule).

Answers Q1 (a) (b) Q2 Q3 (a) (b)

Summary

本章小结

KEY FORMULAS 核心公式

$$n = \frac{N}{N_A}$$

mole 摩尔

$$pV = nRT$$

equation of state

$$pV = NkT$$

molecular form

$$k = \frac{R}{N_A}$$

Boltzmann constant

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

kinetic theory 压强公式

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

mean KE 平均动能

$$c_{rms} = \sqrt{\frac{3kT}{m}}$$

r.m.s. speed

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

two states 两状态

KEY IDEAS 核心思想

- $pV \propto T$ defines the ideal gas; T must be thermodynamic temperature 温度必须用开尔文。
- Pressure arises from molecular bombardment of the walls 压强来自分子撞壁。
- Temperature measures mean molecular kinetic energy 温度即分子平均动能的量度。

Exam Tips

备考提示

- **Always convert to kelvin first** — the single most common error in this topic. 先把温度换成 K。
- Choose the right form: moles $\rightarrow$; molecules $\rightarrow$. Distinguish n , N , m (mass), m (molecule mass). 分清两套符号。
- Derivation marks come in order: momentum change $\rightarrow$ collision rate $\rightarrow$ force $\rightarrow$ pressure. Write each step explicitly. 压强推导按步给分。
- $\langle c^2 \rangle$ is the mean of squares — never . Quote , from the data sheet. 方均 $\neq$ 均方。
- Non-SI units (cm^3 , atm, $^{\circ}\text{C}$) must be converted before substituting. 代入前统一 SI 单位。